

REVIEW ARTICLE

WILEY


**International Journal of
Laboratory Hematology**

New challenges and best practices for the laboratory monitoring of factor VIII and factor IX replacement

D. Van den Bossche¹ | K. Peerlinck^{2,3,4} | M. Jacquemin^{1,2,3,4} 

¹Department of Laboratory Medicine, University Hospitals of Leuven, Leuven, Belgium

²Center for Molecular and Vascular Biology, University of Leuven, Leuven, Belgium

³Department of Vascular Medicine and Hemostasis, University Hospitals Leuven, Leuven, Belgium

⁴National Coordinating Hemophilia Center, University Hospitals Leuven, Leuven, Belgium

Correspondence

Marc Jacquemin, Department of Laboratory Medicine, University Hospitals of Leuven, Leuven, Belgium.
Email: marc.jacquemin@med.kuleuven.be

Abstract

Several recombinant factor VIII and factor IX concentrates with extended half-life (EHL) have recently been validated by clinical studies. The availability of these novel concentrates is expected to significantly facilitate the treatment of patients with hemophilia A and B. However, the modification applied to these molecules has introduced variations in their activity measurement in routine coagulation assays. Depending on the assays, underestimations of up to 10-fold or overestimations of up to approximately 30-fold in the measurements of the recovery have been reported in some factor concentrates. Such biases in monitoring may lead to major under- or overtreatment, as well as unnecessary searching for inhibitor antibodies. In this review, we discuss the guidelines and recommendations that allow the selection of optimal strategies to monitor patients treated with these novel factor concentrates. Based on the specificities of the assays and on local regulations, different chromogenic substrate assays in addition to one-stage clotting assays may be validated to allow the accurate measurement of all novel products. An efficient communication between the clinical laboratory and the clinicians is essential to ensure that the appropriate assays are carried out in laboratories and that the clinicians correctly evaluate the data. Further laboratory and clinical studies are still required for the optimization of the laboratory assays that can be used in the measurement of novel factor VIII and factor IX concentrates with EHL.

KEYWORDS

factor IX, factor VIII, hemophilia

1 | INTRODUCTION

Hemophilia A and B are congenital bleeding disorders characterized by a defect in the factor VIII (FVIII) or factor IX (FIX) genes. Both disorders lead to a deficient or dysfunctional FVIII or FIX protein. Replacement therapy with plasma-derived or recombinant FVIII (rFVIII) and FIX (rFIX) concentrates is the standard of treatment for patients with hemophilia A and B, respectively. Several rFVIII and rFIX concentrates with extended half-life (EHL) have recently been validated in clinical studies. Three EHL rFVIII have been developed by coupling rFVIII with polyethylene glycol. BAY 94-9027 (Bayer, Leverkusen, Germany)¹ and N8-GP (Novo Nordisk, Bagsværd,

Denmark)² are B-domain-deleted (BDD) rFVIII, whereas Bax 855 (Adynovate[®]; Shire, Lexington, MA, USA) is a full-length (FL) rFVIII.³ A pegylated (PEG) rFIX, N9-GP, has been developed by Novo Nordisk.⁴ The fusion of albumin, or the fragment crystallizable (Fc) region of an IgG-molecule with proteins is another strategy that lead to fusion products with EHL lives. Two Fc-fusion products have been developed by Biogen (Cambridge, MA, USA), a BDD rFVIII, rFVIII-Fc (Elocta[®],⁵) and a rFIX, rFIX-Fc (Alprolix[®]).⁶ A fusion product of albumin and rFIX, rIX-FP (Idelvion[®]), has been developed by CSL Behring (Marburg, Germany).⁷ Finally, a B-domain-truncated single-chain rFVIII concentrate, ScrFVIII (Afstyla[®]), has recently been licenced by CSL Behring.⁸

The availability of these novel concentrates is expected to significantly facilitate the treatment of patients with hemophilia A and B. However, the modification applied to these molecules has introduced variations in their activity measurement in routine coagulation assays. Accordingly, the monitoring of patients treated with these novel concentrates must be carried out with carefully selected assays. To appreciate the challenge presented by the monitoring of patients treated with the novel EHL concentrates for laboratories and clinicians, it is important to keep in mind that the measurement of recombinant coagulation factor concentrates has always been complicated by discordance between the measurements carried out with different types of assays. When FL and BDD rFVIII became available, several groups reported differences between FVIII:C levels measured with different types of assays. In 1995, such observations led Jane Lusher and colleagues to emphasize these hurdles with the provocative title: "Will the real FVIII please stand up?"⁹ This question was based on the observation that the levels of a FL rFVIII measured with a chromogenic substrate assay (CSA) were twice as high as those measured with a one-stage clotting assay (OSA) using kaolin as an activator.⁹ Subsequently, important discordances in the measurement of the plasma levels of the BDD rFVIII, ReFacto[®], with different assays led to the recommendation of using a product-specific standard to calibrate the monitoring assays.¹⁰

Although clinical laboratories have been accustomed to discrepancies in the measurement of recombinant factor concentrates for a long time, they need support to be able to deal with the heterogeneity in the measurement of novel EHL products. Depending on the assays, underestimations of up to 10-fold or overestimations of up to about 30-fold in the measurements of the activity of the factors have been reported. Such biases in the monitoring may lead to major under- or overtreatment and to an unnecessary search for inhibitor antibodies. In this review, we therefore discuss the guidelines and recommendations that allow the selection of optimal strategies to monitor patients treated with the novel EHL rFVIII and rFIXs.

2 | GUIDELINES

2.1 | Guidelines for potency assignment

In order to facilitate the implementation of valid monitoring assays in clinical laboratories, it is important to highlight the methods that must be followed by manufacturers to define the potency of new products on the basis of the Scientific and Standardization Committee of the International Society on Thrombosis and Haemostasis (ISTH SSC) recommendations¹¹ and of the national authorities' guidelines.

Prior to the selection of a method for potency labeling of a novel rFVIII or rFIX concentrate, the manufacturer must determine the properties of the new product with a CSA and OSAs using different APTT reagents.¹¹ On the basis of these data, a method for potency labeling is selected in agreement with local regulators. In the European Union

(EU), potency assignment for rFVIII concentrates must be performed preferably with the European Pharmacopoeia FVIII CSA,¹² whereas in the United States, potency labeling can be carried out with OSAs. The European Medicines Agency (EMA) selected the FVIII CSA rather than OSA because the protocol and the reagents of the former could be better standardized than the latter, which can be carried out with different types of activator.¹³ For FIX concentrates, OSAs are recommended, but the interest in FIX CSAs is increasing due to the artefactual activation of PEG rFIX with the silica activator of certain OSAs.

The major clinical risk linked to potency assignment is that the levels determined with a particular assay would not correlate with the activity *in vivo*. For this reason, several types of assays must be carried out to evaluate the activity of the novel EHL products. When FVIII:C levels vary significantly depending on the assays used, it is recommended to demonstrate that the assay used for labeling potency provides results that are comparable to a nonmodified, licenced product.¹³ Importantly, the assays used during the evaluation of the tests for the potency labeling must be calibrated with an international standard concentrate. However, the SSC guidelines recommend carrying out parallel assays calibrated with a plasma standard. The objective of the latter experiments was to determine whether plasma can be used as a reference material to calibrate the assays used in clinical laboratories.

2.2 | Clinical studies

During pharmacokinetics trials, the same assay should preferably be used to assign the potency of the product as to analyze the patient's plasma.¹³ Although apparently obvious, the application of this recommendation does not necessarily prevent biases in the measurement of the recovery. Indeed, the assays used for potency assignment must be calibrated with factor concentrates diluted in factor-deficient plasma, whereas the assays used for postinfusion monitoring are calibrated with normal pooled plasma. These differences in the calibration procedure themselves may lead to a difference in the measurements.¹⁴ In addition, during the pharmacokinetic studies, the effects of using different methods and calibration materials must be evaluated. On the basis of these experiments, the manufacturer determines a dosing guideline, which must allow the clinicians to monitor FVIII recovery by comparing the results from the laboratory with the dose administered.¹¹ The manufacturer must mention in the summary of product characteristics (SmPCs) when discrepant results can be achieved, depending on the assay used for monitoring.¹⁵

2.3 | Postinfusion testing in clinical laboratories

The FVIII and FIX ISTH SSC have stated that in-house assay reagents can be used, provided that these tests are included within the manufacturer's dosing guideline instructions.¹¹ Calibration with the same material as that infused into the patient is recommended but not mandatory unless, according to the manufacturer's dosing guideline,

valid assays cannot be calibrated using the conventional local standard material. The subcommittee further recommended the use of a robust assay and the analysis of several sample dilutions.¹¹ Due to the heterogeneity of the recovery with different assays, the manufacturer's guidelines must identify the preferred type(s) of assay. As practical reasons often limit the use of the preferred assay(s), the manufacturer also indicates the alternative methods that are merely acceptable.

Given the diverse combination of reagents and automates, comparative studies can also be carried out by organizations, independent of established companies. The difficulty in these studies lies in the definition of acceptable bias, which is typically chosen based on the variability of FVIII and FIX assays in international proficiency testing. As the interlaboratory variability of these tests is generally quite large, the maximum bias that has been considered as acceptable was as high as 30%.¹⁶ Although such studies may play an important role in identifying useful assays, their conclusions are not necessarily reflected in the manufacturer's dosing guidelines. Similarly, individual laboratories may wish to validate a local assay not previously evaluated by the manufacturer. These occurrences have not been considered in ISTH SSC recommendations, which creates significant uncertainty for local laboratories that consider using these reagents. This situation is further complicated by the limited data available, which prevent independent organizations or authorities from formally recommending the use of any reagent.¹⁷

3 | DISCORDANCE BETWEEN FVIII AND FIX LEVELS MEASURED WITH OSA AND CSA

Many studies have been carried out to evaluate the suitability of different assays for the monitoring of EHL rFVIII and rFIX concentrates. The available data describing the interactions between these products and OSAs or CSAs are summarized in Tables 1-3.

3.1 | EHL rFVIII

Within the normal range, the ScrFVIII (Afstyla[®]) levels measured with OSAs using silica or ellagic acid as activator reach only 40%-60% of the expected recovery, whereas the concentrations measured with the CSAs are close to the target values (Table 1).¹⁸ Accordingly, the company recommends either the use of a CSA for the measurement of ScrFVIII postinfusion levels, or, if the ScrFVIII levels are measured with an OSA, a multiplication of the results by a correction factor of 2.¹⁸ This strategy has been validated by the Food and Drug Administration. In the low range, the levels determined with OSA are closer to the target value, so that a multiplication of the FVIII:C activity by the correction factor results in a twofold overestimation (Table 1).¹⁸ By contrast, the postinfusion levels of rFVIIIIFc (Elocta[®]) can be measured just as acceptable with the OSAs as with the CSAs (Table 1).¹⁹

Interestingly, the interactions between the 3 PEG rFVIII and the OSAs reagents are significantly different. The recoveries of BAY 94-9027 achieved with OSAs are very discrepant when viewed as a function of the assays. Using OSAs with silica as an activator, the FVIII:C levels are underestimated 10-fold or are unmeasurable as the coagulation curves are grossly abnormal, whereas a proper recovery is determined when using OSA activated with ellagic acid.²⁰ By contrast, the bias in the measurement of the N8-GP levels is less pronounced, although a twofold underestimation of the expected recovery is observed with some reagents (Table 2).²¹ Importantly, unacceptable results were reported when a reagent was used on 1 type of coagulation automate but not another. Notably, the OSA reagents SynthAsil and SynthAFax provided accurate results when used on the coagulation automate ACL TOP according to the manufacturer's methods, but not on other instruments with protocols developed locally.²¹ This observation has important implications for the selection and validation of OSAs by clinical laboratories.

For the BAX 855, Adynovate[®], expected recoveries are measured within the normal range for all OSAs.^{22,23} In the low range, 1 study reported recoveries close to the expected values with all assays,²² whereas another reported up to a threefold overestimation in the low range.²³ In the latter study, a product-specific standard was also used, which resulted in a better evaluation of the recovery but did not fully prevent the overestimation in the low range.²³ None of these studies provided data on the individual reagents, so the possibility of more severe biases with 1 reagent or the other cannot be fully excluded. In contrast to the discrepancies reported with the OSAs, the levels of the 3 PEG rFVIII can be measured with CSAs (Table 1).²⁰⁻²⁴ However, results generated with the CSA must be analyzed with caution, as 1 recent study reported important under- and overestimations of the BAX 855, Adynovate[®], levels with this type of assay,²³ while another study highlighted a higher interlaboratory variability compared to the OSA, especially with low FVIII concentrations.²²

3.2 | EHL rFIX

The levels of the PEG rFIX, N9-GP, are overestimated between five-fold and 30-fold with silica-based assays, whereas with a reagent containing very small particles of silica (called colloidal silica) such as the SynthAsil[®] (Werfen), or with reagents containing ellagic acid, recoveries range between 30% and 60% of the expected value.^{25,26} Limited data are available for the rIX-albumin, as most studies have been carried out with Pathromtin SL, a silica-based reagent.^{7,27} The levels of rFIX-FP were severely underestimated with 1 ellagic and 1 kaolin-based OSA reagent. Other OSAs are reported as potentially useful, but the data have only been published as an abstract.^{7,27} The measurement of rFIXFc can be carried out with all assays tested so far, with the exception of a kaolin-based assay, with which the levels are underestimated approximately twofold.¹⁴ Interestingly, the levels of all types of EHL rFIX can be accurately measured with CSAs.^{7,14,25-27}

TABLE 1 Measurement of ScrFVIII and rFVIII Fc with OSA and CSA

Reagent	Assay type	Activator	Percentage of FVIII:C expected recovery (%) ^a			
			ScrFVIII Afstyl ^a (BDD)	rFVIII Fc Elocta (BDD)	Normal FVIII levels ^b	Low FVIII levels ^c
SynthASil	OSA	Colloidal silica	39 ¹⁸ with CF: 77 ¹⁸	100 ¹⁸ with CF: 200 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
APTT-SP	OSA	Silica	52 ¹⁸ with CF: 104 ¹⁸	100 ¹⁸ with CF: 200 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
STA ^R -PTT Automate	OSA	Silica	n.d.	n.d.	95 ± 15 ¹⁹	116 ± 36 ¹⁹
Pathromtin SL	OSA	Silica	41 ¹⁸ with CF: 82 ¹⁸	100 ¹⁸ with CF: 200 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
TriniClot aPTT S	OSA	Silica	n.d.	n.d.	n.d.	n.d.
Trinity Auto APTT	OSA	Silica	n.d.	n.d.	n.d.	n.d.
STA ^R -C.K. Prest ^R	OSA	Kaolin	n.d.	n.d.	95 ± 15 ¹⁹	116 ± 36 ¹⁹
STA ^R -Cephascreen ^R	OSA	Polyphenols	n.d.	n.d.	95 ± 15 ¹⁹	116 ± 36 ¹⁹
Actin	OSA	Ellagic acid	n.d.	n.d.	95 ± 15 ¹⁹	116 ± 36 ¹⁹
Actin FS	OSA	Ellagic acid	56 ¹⁸ with CF: 113 ¹⁸	150 ¹⁸ with CF: 300 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
Actin FSL	OSA	Ellagic acid	57 ¹⁸ with CF: 114 ¹⁸	100 ¹⁸ with CF: 200 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
SynthAFax	OSA	Ellagic acid	58 ¹⁸ with CF: 116 ¹⁸	200 ¹⁸ with CF: 400 ¹⁸	95 ± 15 ¹⁹	116 ± 36 ¹⁹
DG Synth	OSA	Ellagic acid	n.d.	n.d.	n.d.	n.d.
Siemens FVIII chromogenic	CSA	-	n.d.	n.d.	119 ± 23 ¹⁹	120 ± 37 ¹⁹
Biophen FVIII:C	CSA	-	113 ¹⁸	150 ¹⁸	119 ± 23 ¹⁹	120 ± 37 ¹⁹
Coamatic FVIII	CSA	-	Test used in clinical study ³⁹	Test used in clinical study ³⁹	119 ± 23 ¹⁹	120 ± 37 ¹⁹
Coatest SP	CSA	-	n.d.	n.d.	119 ± 23 ¹⁹	120 ± 37 ¹⁹
Electachrome FVIII	CSA	-	n.d.	n.d.	n.d.	n.d.
Technochrom FVIII:C	CSA	-	n.d.	n.d.	119 ± 23 ¹⁹	120 ± 37 ¹⁹

^aThe percentages of expected recovery are calculated as (measured FVIII:C levels/recovery expected according the labeled potency of the concentrate spiked into factor-deficient plasma or to the levels determined with a reference assay) × 100. Depending on the available data, the percentages of recovery are expressed as mean or median, mean ± SD, or range.

^{b,c}Spiked samples or patients' plasma containing rFVIII levels between 70 and 130 and ≤5 IU/dL, respectively.

^{d,e}With ACL-TOP and with other instrument(s), respectively.

CF: correction factor of 2, the use of that is recommended by the company when BDD ScrFVIII levels are measured with OSA; n.d.: no data available for this reagent.

Bold numbers highlight percentages of recovery <70% (underestimation of FVIII levels by more than 30%) and >130% (overestimation of FVIII levels by more than 30%), which are considered as unacceptable for the monitoring of patients treated with FVIII concentrates.¹⁶ Percentages of recovery in italics are from studies that did not provide data for the individual reagents.

4 | MECHANISMS RESPONSIBLE FOR DISCREPANCIES BETWEEN FVIII AND FIX LEVELS MEASURED WITH OSA AND CSA

Besides the modification of the rFVIII and rFIX molecules specifically aimed at prolonging the half-life, several characteristics of the unmodified molecules themselves, as well as of the assays, contribute to the discrepancies in the measured recoveries. In addition, the

methods used to label the potency of the rFVIII and of the rFIX concentrates rely on calibration with factor concentrates, whereas in clinical laboratories, the assays are calibrated with plasma. By itself, the use of different calibrators can lead to a 10%-20% difference in recovery.¹⁴

Differences in the recovery of FL rFVIII with different types of assay have been recognized for a long time, although, thus far, the mechanisms responsible for these discrepancies have not been

TABLE 2 Measurement of pegylated rFVIII with OSA and CSA

Reagent	Percentage of FVIII:C expected recovery (%) ^a					
	BAY 94-9027 (BDD)		N8-GP (BDD)		BAX 855 Adynovate (FL)	
	Normal FVIII levels ^b	Low FVIII levels ^c	Normal FVIII levels ^b	Low FVIII levels ^c	Normal FVIII levels ^b	Low FVIII levels ^c
SynthASil	n.d.	n.d.	127-133 ^d 57-75 ^{e21}	n.d.	96-107 ²²	117-133 ²²
APTT-SP	10²⁰	10²⁰	45-55²¹	n.d.	96-107 ²² 97-159 ²³	117-133 ²² 165-262²³
STA-PTT A	Very low recovery²⁰	Very low recovery²⁰	n.d.	n.d.	96-107 ²²	117-133 ²²
Pathromtin SL	n.d.	n.d.	95-117 ²¹	n.d.	97-159 ²³	165-262²³
TriniClot aPTT S	n.d.	n.d.	n.d.	n.d.	96-107 ²²	117-133 ²²
STA-CK Prest	n.d.	n.d.	n.d.	n.d.	96-107 [22] 97-159 ²³	117-133 ²² 165-262²³
Cephascreeen	112 ± 1 ²⁰	100 ± 20 ²⁰	n.d.	n.d.	96-107 ²²	117-133 ²²
Actin	CT as with plasma ²⁰	CT as with plasma ²⁰	79-82 ²¹	n.d.	n.d.	n.d.
Actin FS	n.d.	n.d.	72-87 ²¹	n.d.	96-107 ²²	117-133 ²²
Actin FSL	n.d.	n.d.	70-80 ²¹	n.d.	96-107 ²² 97-159 ²³	117-133 ²² 165-262²³
SynthAFax	CT as with plasma ²⁰	CT as with plasma ²⁰	75-80 ^d 55-80 ^{e21}	n.d.	n.d.	n.d.
DG Synth			60-74²¹	n.d.	n.d.	n.d.
Siemens FVIII chromogenic	n.d.	n.d.	100-134 ²¹ 113 ± 16 ²⁴	113 ± 16 ²⁴	144-221²³	3-290²³
Biophen FVIII:C	110 ± 7 ²⁰	80 ± 20 ²⁰	125-129 ²¹ 118 ± 6 ²⁴	158 ± 9²⁴	115-134 ²² 144-221²³	76-120 ²² 3-290²³
Coamatic FVIII	n.d.	n.d.	120-156 ²¹ 125 ± 4 ²⁴	109 ± 6 ²⁴	115-134 ²²	76-120 ²²
Coatest SP	n.d.	n.d.	92-143 ²¹ 110 ± 2 ²⁴	103 ± 4 ²⁴	n.d.	n.d.
Electachrome FVIII	n.d.	n.d.	n.d.	n.d.	115-134 ²² 144-221²³	76-120 ²² 3-290²³
Technochrom FVIII:C	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.

^aThe percentages of expected recovery are calculated as (measured FVIII:C levels/recovery expected according the labeled potency of the concentrate spiked into factor-deficient plasma or to the levels determined with a reference assay) × 100. Depending on the available data, the percentages of recovery are expressed as mean or median, mean ± SD, or range.

^{b,c}Spiked samples or patients' plasma containing rFVIII levels between 70 and 130 and ≤5 IU/dL, respectively.

^{d,e}With ACL-TOP and with other instrument(s), respectively.

CT: clotting time measured with OSA; n.d.: no data available for this reagent. Bold numbers highlight percentages of recovery <70% or >130%, which are considered as unacceptable for the monitoring of patients treated with FVIII concentrates.¹⁶ Percentages of recovery in italics are derived from studies that did not provide data for the individual reagents.

Bold numbers highlight percentages of recovery <70% (underestimation of FVIII levels by more than 30%) and >130% (overestimation of FVIII levels by more than 30%), which are considered as unacceptable for the monitoring of patients treated with FVIII concentrates.¹⁶ Percentages of recovery in italics (bold or plain) are from studies that did not provide data for the individual reagents.

clearly described. The reasons why the discrepancies with some FL rFVIII are more pronounced than with others are also still unknown. For the BDD rFVIII ReFacto^R and ReFacto AF^R, the difference between the recovery with OSA and CSA can be up to 50%.^{10,28-30} In this case, the measurement of the product was shown to be highly dependent on the type and concentration of the phospholipids used in the assay.³¹ However, the mechanisms may be more complex, as

the discordance between the assays also varies from 1 patient to another.³²

In addition to the discordances reported with FL and BDD rFVIII, the modifications applied to extend the half-life of rFVIII and of rFIX have created novel interactions with the reagents of the coagulation assays. Several mechanisms responsible for the discordant measurement of EHL rFVIII and of rFIXs have been

TABLE 3 Measurement of N9-GP, rIX-FP, and rFIXFc with OSA and CSA

Reagent	Percentage of FIX:C expected recovery (%) ^a					
	N9-GP		rIX-FP Idelvion ^R		rFIXFc Alprolix ^R	
	Normal FVIII levels ^b	Low FVIII levels ^c	Normal FVIII levels ^b	Low FVIII levels ^c	Normal FVIII levels ^b	Low FVIII levels ^c
SynthASil	62 ^d 39-48 ^{e25}	185 ^d 37-74 ^{e25}	n.d. ²⁷	n.d.	63-128 ¹⁴	100-200 ¹⁴
APTT-SP	536-1247 ²⁵	333-778 ²⁵	n.d. ²⁷	n.d.	79 ¹⁴	n.d.
STA ^R -PTT Automate	n.d.	n.d.	n.d. ²⁷	n.d.	58-99 ¹⁴	80-140 ¹⁴
Pathromtin SL	1550-2811 ²⁵	531-832 ²⁵	Used for clinical studies ⁷	Used for clinical studies ⁷	70-103 ¹⁴	100-120 ¹⁴
TriniClot aPTT S	n.d.	n.d.	n.d. ²⁷	n.d.	n.d. ¹⁴	n.d.
Trinity Auto APTT	n.d.	n.d.	n.d. ²⁷	n.d.	100-129 ¹⁴	70-140 ¹⁴
STA ^R -C.K. Prest ^R	n.d.	n.d.	50 ²⁷	n.d.	43-60 ¹⁴	60-104 ¹⁴
STA ^R -Cephascreen ^R	91 ± 13 ²⁶	119 ± 19 ²⁶	n.d. ²⁷	n.d.	72-75 ¹⁴	118-120 ¹⁴
Actin	57 ²⁵	148 ²⁵	n.d. ²⁷	n.d.	105 ¹⁴	n.d.
Actin FS	31-44 ²⁵	74-99 ²⁵	50 ²⁷	n.d.	105-130 ¹⁴	120-220 ¹⁴
Actin FSL	44 ²⁵	74 ²⁵	n.d. ²⁷	n.d.	100-131 ¹⁴	96-360 ¹⁴
SynthAFax	108 ^d 85-102 ^{e25}	248 ^d 111-148 ^{e25}	n.d. ²⁷	n.d.	120 ¹⁴	n.d.
DG Synth	83%-99% ²⁵	148-185 ²⁵	n.d.	n.d.	n.d.	n.d.
ROX factor IX CSA	109-116 ²⁵ 104 ± 10 ²⁶	111-126 ²⁵ 110 ± 10 ²⁶	n.d.	n.d.	n.d.	n.d.
BIOPHEN factor IX CSA	91 ²⁵ 94 ± 5 ²⁶	74 ²⁵ 77 ± 8 ²⁶	n.d.	n.d.	101 ¹⁴	84 ¹⁴

^aThe percentages of expected recovery are calculated as (measured FVIII:C levels/recovery expected according the labeled potency of the concentrate spiked into factor-deficient plasma or to the levels determined with a reference assay) × 100. Depending on the available data, the percentages of recovery are expressed as mean or median, mean ± SD, or range.

^{b,c}Spiked samples or patients' plasma containing rFVIII levels between 70 and 130 and ≤5 IU/dL, respectively.

^{d,e} With ACL-TOP and with other instrument(s), respectively.

n.d.: no data available for this reagent. Bold numbers highlight percentages of recovery <70% or >130%, which are considered as unacceptable for the monitoring of patients treated with FVIII concentrates.¹⁶ Percentages of recovery in italics are from studies that did not provide data for the individual reagents.

Bold numbers highlight percentages of recovery <70% (underestimation of FVIII levels by more than 30%) and >130% (overestimation of FVIII levels by more than 30%), which are considered as unacceptable for the monitoring of patients treated with FVIII concentrates.¹⁶

unambiguously identified for some pairings of reagent/product, though not for all.

4.1 | EHL rFVIIS

The recovery of the PEG rFVIII BAY 94-9027 is underestimated by more than 10-fold with some OSAs activated by silica (Table 1).²⁰ Gu et al²⁰ have attributed the reduced activity of BAY 94-9027 to inhibition of the contact pathway by PEG moieties. The relevance of this explanation is unclear, as the concentrations of PEG required to inhibit the contact factor pathway are higher than those present in FVIII concentrates. It is also plausible that BAY 94-9027 binds to the silica particles through the PEG moieties, thereby preventing activation by thrombin or tenase complex formation. By contrast, the activity of another PEG rFVIII, BAX 855, is not affected by reagents

containing silica. One plausible explanation is that the smaller size of the PEG moiety in this rFVIII makes them less prone to bind to silica. Alternatively, the sites where the PEG moieties are bound to the rFVIII molecules may be determinant. Indeed, BAX 855 is a full-length rFVIII, and a majority of PEG moieties are located in the B domain.³ Accordingly, the BAX 855 molecules may bind to the silica beads through PEG moieties in the B domain but then detach from the immobilized B domain following activation by thrombin. Free activated BAX 855 may then play a normal role in the coagulation processes.

The mechanism by which the single-chain ScrFVIII (Afstyla^R) is underestimated by about twofold by all OSAs has not yet been elucidated. The modified rFVIII does have a higher affinity for VWF⁸; however, upon activation by thrombin, the small fragment of the B domain that provides the covalent bridge between the heavy and

the light chains of the FVIII molecule is cleaved. The structure of the activated ScrFVIII is then identical to the native FVIII.^{8,33} A plausible hypothesis to explain the reduced FVIII activity in OSAs is thus that the rate of ScrFVIII activation by thrombin is slower than for the wild-type FVIII. During CSAs, the incubation of FVIII with thrombin is longer, which may render this type of assay more suitable for the measurement of ScrFVIII levels.

4.2 | EHL rFIX

The mechanism responsible for the overestimation of PEG rFIX N9-GP activity when using most silica-based OSAs calibrated with plasma FIX has recently been established.³⁴ N9-GP is adsorbed by the silica particles through the PEG moieties. It is then activated by plasma kallikrein and by activated factor XI during the incubation of the citrated plasma sample with silica. When Ca^{++} is added to the reaction, the activated N9-GP accelerates the formation of the clot, leading to a more than 10-fold overestimation of FIX:C levels. Interestingly, this artefactual activation of N9-GP is not observed when SynthASil^R, a reagent containing silica in the form of colloidal particles, is used as an activator for the OSA. N9-GP does not bind to the colloidal silica particles, either because of the properties of the particles or because of the small quantity of silica in this reagent.³⁴

Limited data are available for the rIX-FP (Idelvion^R) as most studies have been carried out with Pathromtin SL, a silica-based reagent. The mechanism leading to the underestimation of the rIX-FP levels with some ellagic and kaolin OSA reagents is still unknown.²⁷ Similarly, it is still unclear why the measurement of rFIXFc (Alprolix^R) can be accurately carried out with all assays tested so far, except when kaolin is used as an activator of the OSA.¹⁴

5 | HOW CAN WE DO BETTER?

5.1 | CSAs?

Chromogenic substrate assay is frequently considered to be the assay of choice for monitoring patients treated with several EHL rFVIII or rFIX concentrates. However, one limitation in the use of this type of assay is that it is currently available only in a small number of laboratories. This is somewhat unfortunate, as in addition to its potential usefulness for the monitoring of FVIII replacement therapy, the FVIII CSA is necessary to diagnose some moderate hemophilia A variants, due to mutations that reduce the stability of the activated FVIII molecule. Such mutations can be detected in this assay due to a prolonged incubation period that allows the A2 domain to dissociate from the activated FVIII molecule.^{35,36} Accordingly, a broader availability of the FVIII CSA may be useful not only for monitoring but also for the diagnosis of certain patients with hemophilia A. One potential limitation of this type of assay that deserves further investigation is the recently reported higher interlaboratory variability when compared with the OSA, especially at low FVIII concentrations.²²

5.2 | Correction factors?

Recent recommendations by the UKHCDO discouraged the use of assays requiring mathematical corrections.¹⁷ Potential concerns over such a procedure are that it requires faultless communication between the ordering clinicians and the laboratory. Furthermore, it remains to be determined whether an identical underestimation of FVIII activity with OSAs is observed with all patients and in all circumstances, notably during surgical intervention. By comparison, we recently observed that in samples from patients treated with ReFacto AF^R, the ratios between FVIII:C levels measured with a CSA and with an OSA varied from 1 patient to another, despite both assays having been calibrated with the product-specific calibrator, ReFacto AF laboratory standard.³² As all samples analyzed during that study were collected during and immediately after a surgical intervention, it cannot be excluded that the ratios between the OSAs and the CSAs were not only dependent on the patients themselves but also on the type of surgical intervention. In addition, batch-to-batch variation, most notably in the concentration of phospholipids, may contribute to discrepancies between assays.³⁷ More data from samples collected in different circumstances are thus required to validate the use of correction factors.

5.3 | Spiked control material?

The use of a control material prepared by spiking factor-deficient plasma with EHL products can significantly aid the validation of the assays. Such control materials can be prepared in local laboratories by diluting commercially available EHL factor vials. However, the dilution process itself may be a cause of variability. A better method in terms of reproducibility may be achieved if companies provide the laboratories with control plasma samples spiked with a factor level determined by a reference method. However, deficient plasma spiked with EHL products can only be used as a valid control if the commutability of this material with ex vivo samples has been demonstrated.³⁸ Indeed, concentrates may contain significant activated, misfolded or aggregated proteins that are rapidly cleared from plasma upon administration. In such cases, the activity of proteins remaining in plasma may be different to that of controls. Interestingly, the discrepancy in the measurement of EHL products with different assays can be a useful parameter in demonstrating their commutability. Spiked samples showing the same discordances between assays as plasma samples from treated patients are likely highly representative of the latter.³⁸ The evaluation of the commutability of spiked samples must also take into account potential variability from patient to patient.³²

5.4 | Adaptation of the reportable range?

A recurring observation in the published data analyzing spiked samples with EHL coagulation factors is the overestimation of FVIII activity in the low range. This overestimation is not due to the properties of the modified coagulation factors themselves as it is

also observed with control recombinant factors such as Advate^R or Benefix^R.^{14,19} It is unlikely that this bias is due only to the poor reproducibility of the assays, as then a clear trend toward overestimation would not be observed. Other factors, such as the generation of the calibration curve in diluent buffers or in factor-deficient plasma, and the use (or not) of a high- and low-range calibration curve may play some part. Only limited data are available to determine whether the use of product-specific standards would improve the accuracy of measurements in the low range.²³

The presence of such a bias may require the adaptation of the reportable range,²⁶ especially to avoid giving false confidence in the trough levels achieved with novel EHL concentrates. Conversely, in rare cases, such as that of the measurement of rFIXFc with an OSA using kaolin as activator,¹⁴ some assays are more accurate in the low range than in the high range. Good communication with clinicians is therefore crucial to select the appropriate assays as a function of the product. Conversely, the clinician must be aware of the potential biases of the assay, particularly in the context of surgical intervention or when worrying about the presence of an inhibitor.

6 | CONCLUSIONS

The discordance in the levels of EHL rFVIII and rFIXs measured with different assays is a challenge for clinical laboratories, as well as for the clinicians monitoring treatment with these novel drugs. Based on the specificities of the assays and on local regulations, various methods—including CSAs in addition to OSAs—may be validated to facilitate the accurate measurement of all novel EHL products and the correct diagnosis of the various types of hemophilia A or B. Accordingly, efficient communication between the laboratory and the clinicians is essential to guarantee that the appropriate assays are carried out in the laboratories and that the clinicians correctly evaluate the data. The companies developing the novel EHL factor concentrates play a key role in supporting the correct monitoring of their product by providing relevant information to both local laboratories and clinicians. In addition, they can significantly contribute to the validation of appropriate methods in clinical laboratories and the organization of proficiency testing by providing quality control material with specified concentrations of their products. Finally, although methods for the potency labeling of coagulation factor concentrates by the manufacturers have been precisely defined by local authorities and the ISTH SSC, further guidance is still expected by the clinical laboratories who monitor patients treated with these products. Further laboratory and clinical studies are also required for the optimization of the laboratory assays that can be used for the measurement of novel EHL rFVIII and rFIXs.

ORCID

M. Jacquemin  <http://orcid.org/0000-0002-4780-268X>

REFERENCES

1. Reding MT, Ng HJ, Poulsen LH, et al. Safety and efficacy of BAY 94-9027, a prolonged-half-life factor VIII. *J Thromb Haemost.* 2017;15:411-419.
2. Tiede A, Brand B, Fischer R, et al. Enhancing the pharmacokinetic properties of recombinant factor VIII: first-in-human trial of glyco-PEGylated recombinant factor VIII in patients with hemophilia A. *J Thromb Haemost.* 2013;11:670-678.
3. Turecek PL, Bossard MJ, Graninger M, et al. BAX, 855, a PEGylated rFVIII product with prolonged half-life. Development, functional and structural characterisation. *Hämostaseologie.* 2012;32(Suppl 1):S29-S38.
4. Tiede A, Abdul-Karim F, Carcao M, et al. Pharmacokinetics of a novel extended half-life glycoPEGylated factor IX, nonacog beta pegol (N9-GP) in previously treated patients with haemophilia B: results from two phase 3 clinical trials. *Haemophilia.* 2017;23:547-555.
5. Powell JS, Josephson NC, Quon D, et al. Safety and prolonged activity of recombinant factor VIII Fc fusion protein in hemophilia A patients. *Blood.* 2012;119:3031-3037.
6. Pasi KJ, Fischer K, Ragni M, et al. Long-term safety and efficacy of extended-interval prophylaxis with recombinant factor IX Fc fusion protein (rFIXFc) in subjects with haemophilia B. *Thromb Haemost.* 2017;117:508-518.
7. Santagostino E, Negrier C, Klamroth R, et al. Safety and pharmacokinetics of a novel recombinant fusion protein linking coagulation factor IX with albumin (rIX-FP) in hemophilia B patients. *Blood.* 2012;120:2405-2411.
8. Schmidbauer S, Witzel R, Robbel L, et al. Physicochemical characterisation of rVIII-SingleChain, a novel recombinant single-chain factor VIII. *Thromb Res.* 2015;136:388-395.
9. Lusher JM, Hillman-Wiseman C, Simpson P, Hurst D. Plasma FVIII levels measured after infusion of recombinant FVIII (rFVIII) vary significantly with different assay methods – (Will the real FVIII level please stand up?). *Blood.* 1995;86(Suppl. 1):192a. (abstract 755).
10. Ingerslev J, Jankowski MA, Weston SB, ReFacto field study participants. Collaborative field study on the utility of a BDD factor VIII concentrate standard in the estimation of BDDr Factor VIII: C activity 9 in hemophilic plasma using one-stage clotting assays. *J Thromb Haemost.* 2004;2:623-628.
11. Hubbard AR, Dodt J, Lee T, et al. Recommendations on the potency labelling of factor VIII and factor IX concentrates. *J Thromb Haemost.* 2013;11:988-989.
12. European Directorate for the Quality of Medicines & HealthCare (EDQM), Council of Europe. European Pharmacopoeia, General Notices. In: *European Pharmacopoeia*, 8th edn, Strasbourg, France: European Directorate for the Quality of Medicines & HealthCare (EDQM), Council of Europe; 2015.
13. Committee for medicinal products for human use (CHMP). Guidelines on the Clinical Investigation of Recombinant and Human Plasma-Derived Factor VIII Products; 2016. EMA/CHMP/BPWP/144533/2009 rev. 1
14. Sommer JM, Buyue Y, Bardan S, et al. Comparative field study: impact of laboratory assay variability on the assessment of recombinant factor IX Fc fusion protein (rFIXFc) activity. *Thromb Haemost.* 2014;112:932-940.
15. Committee for Medicinal Products for Human Use (CHMP). Guideline on Core SmPC for Human Plasma Derived and Recombinant Coagulation factor VIII Products; 2017. EMA/CHMP/BPWP/1619/1999 rev. 3.
16. Kitchen S, Tiefenbacher S, Gosselin R. Factor activity assays for monitoring extended half-life FVIII and factor IX replacement therapies. *Semin Thromb Hemost.* 2017;43:331-337.

17. Collins P, Chalmers E, Chowdary P, et al. The use of enhanced half-life coagulation factor concentrates in routine clinical practice: guidance from UKHCDO. *Haemophilia*. 2016;22:487-498.
18. Bowyer A, Key N, Dalton D, Kitchen S, Makris M. The coagulation laboratory monitoring of Afstyla single-chain FVIII concentrate. *Haemophilia*. 2017;23:e469-e470.
19. Sommer JM, Moore N, McGuffie-Valentine B, et al. Comparative field study evaluating the activity of recombinant factor VIII Fc fusion protein in plasma samples at clinical haemostasis laboratories. *Haemophilia*. 2014;20:294.
20. Gu JM, Ramsey P, Evans V, et al. Evaluation of the activated partial thromboplastin time assay for clinical monitoring of PEGylated recombinant factor VIII (BAY 94-9027) for haemophilia A. *Haemophilia*. 2014;20:593-600.
21. Hillarp A, Bowyer A, Ezban M, Persson P, Kitchen S. Measuring FVIII activity of glycopegylated recombinant factor VIII, N8-GP, with commercially available one-stage clotting and chromogenic assay kits: a two-centre study. *Haemophilia*. 2017;23:458-465.
22. Turecek PL, Romeder-Finger S, Apostol C, et al. A world-wide survey and field study in clinical haemostasis laboratories to evaluate FVIII: C activity assay variability of ADYNOVATE and OBIZUR in comparison with ADVATE. *Haemophilia*. 2016;22:957-965.
23. Bulla O, Poncet A, Alberio L, et al. Impact of a product-specific reference standard for the measurement of a PEGylated rFVIII activity: the Swiss Multicentre Field Study. *Haemophilia*. 2017;23:e335-e339.
24. Pickering W, Hansen M, Kjalke M, Ezban M. Factor VIII chromogenic assays can be used for potency labeling and postadministration monitoring of N8-GP. *J Thromb Haemost*. 2016;14:1579-1587.
25. Bowyer AE, Hillarp A, Ezban M, Persson P, Kitchen S. Measuring factor IX activity of nonacog beta pegol with commercially available one-stage clotting and chromogenic assay kits: a two-center study. *J Thromb Haemost*. 2016;14:1428-1435.
26. Tiefenbacher S, Bohra R, Amiral J, et al. Qualification of a select one-stage activated partial thromboplastin time-based clotting assay and two chromogenic assays for the post-administration monitoring of nonacog beta pegol. *J Thromb Haemost*. 2017;15:1901-1912.
27. St Ledger K, Fuessner A, Kalina U, Jacobs I, Voigt C, Bensen-Kennedy D. Performance of a recombinant fusion protein linking coagulation factor IX with albumin (rFIX-FP) in the one-stage assay. *Haemophilia*. 2016;22(Suppl 4):60. (Abstract)
28. Morfini M, Cinotti S, Bellatreccia A, Paladino E, Gringeri A, ReFacto-AICE Study Group. A multicenter pharmacokinetic study of the B-domain deleted recombinant factor VIII concentrate using different assays and standards. *J Thromb Haemost*. 2003;1:2283-2289.
29. Cauchie M, Toelen J, Peerlinck KM, Jacquemin M. Practical and cost-effective measurement of B-domain deleted and full-length recombinant FVIII in the routine haemostasis laboratory. *Haemophilia*. 2013;19:e133-e138.
30. Pouplard C, Ternisien C, Desconclois C, Lasne D, Aillaud MF, Caron C. Discrepancies between one stage assay and chromogenic substrate assay in patients treated with recombinant or plasma-derived FVIII and usefulness of a specific standard in ReFacto AF^R-treated patients. *Haemophilia*. 2016;22:e101-e103.
31. Mikaelsson M, Oswaldsson U, Sandberg H. Influence of phospholipids on the assessment of factor VIII activity. *Haemophilia*. 1998;4:646-650.
32. Jacquemin M, Vodolozkaia A, Toelen J, et al. Measurement of B-domain-deleted ReFacto AF activity with a product-specific standard is affected by choice of reagent and patient-specific factors. *Haemophilia*. 2017. <http://doi.org/10.1111/hae.13123> [Epub prior to print]
33. Zollner S, Raquet E, Claar P, et al. Non-clinical pharmacokinetics and pharmacodynamics of rVIII-SingleChain, a novel recombinant single-chain factor VIII. *Thromb Res*. 2014;134:125-131.
34. Rosén P, Rosén S, Ezban M, Persson E. Overestimation of N-glycoPEGylated factor IX activity in a one-stage factor IX clotting assay owing to silica-mediated premature conversion to activated factor IX. *J Thromb Haemost*. 2016;14:1420-1427.
35. Parquet-Gerneiz A, Mazurier C, Goudemand M. Functional and immunological assays of FVIII in 133 haemophiliacs—characterization of a subgroup of patients with mild haemophilia A and discrepancy in 1- and 2-stage assays. *Thromb Haemost*. 1988;59:202-206.
36. Duncan EM, Duncan BM, Tunbridge LJ, Lloyd JV. Familial discrepancy between the one-stage and two-stage factor VIII methods in a subgroup of patients with haemophilia A. *Br J Haematol*. 1994;87:846-848.
37. Kitchen S, Cartwright I, Woods TA, Jennings I, Preston FE. Lipid composition of seven APTT reagents in relation to heparin sensitivity. *Br J Haematol*. 1999;106:801-808.
38. Sørensen MH, Andersen S, Ezban M. Factor IX-deficient plasma spiked with N9-GP behaves similarly to N9-GP post-administration clinical samples in N9-GP ELISA and FIX activity assays. *Haemophilia*. 2015;21:832-836.
39. Zhang Y, Roberts J, Tortorici M, et al. Population pharmacokinetics of recombinant coagulation factor VIII-SingleChain in patients with severe hemophilia A. *J Thromb Haemost*. 2017;15:1106-1114.

How to cite this article: Van den Bossche D, Peerlinck K, Jacquemin M. New challenges and best practices for the laboratory monitoring of factor VIII and factor IX replacement. *Int J Lab Hem*. 2018;40(Suppl. 1):21-29. <https://doi.org/10.1111/ijlh.12813>